HL-50E Series - Installation

AD0877

Revision E - 07/2026

Installation Requirements

This procedure will guide you through the installation of a HL-50E and HL-50E-XT.
The fiber laser marking machine is composed of three main components:

- Laser Source System
- Galvonometer Scanning System
- Desktop/Windows 11 Operating System

Tools Required:

- ✓ Forklift
- ✓ Hexkey Set *(included)*
- ✓ Phillips Screwdriver
- ✓ Flat Head Screwdriver
- ✓ Internet
- ✓ Dark Colored Construction Paper (Black is recommended for best results)
- ✓ Caliper

Windows 11 Operating System:

⚠ Haas Automation provides a Windows 11 Pro license with the purchase of a HL50E Fiber Laser Marking Machine. All technical support, troubleshooting, or assistance related to the installation, use, or maintenance of the Windows 11 Pro software needs to be directed to Microsoft or a qualified IT professional.

To learn how to activate ***Windows 11 Pro*** *using a product key CLICK HERE.*

LightBurn Software Activation:

To learn how to activate LightBurn *CLICK HERE*, or refer to the **LightBurn License Activation** section below.

⚠ **IMPORTANT: Internet will be needed to complete the Lightburn installation.**

For additional information on how to get started with LightBurn *CLICK HERE*.

To see LightBurn tutorials *CLICK HERE*.

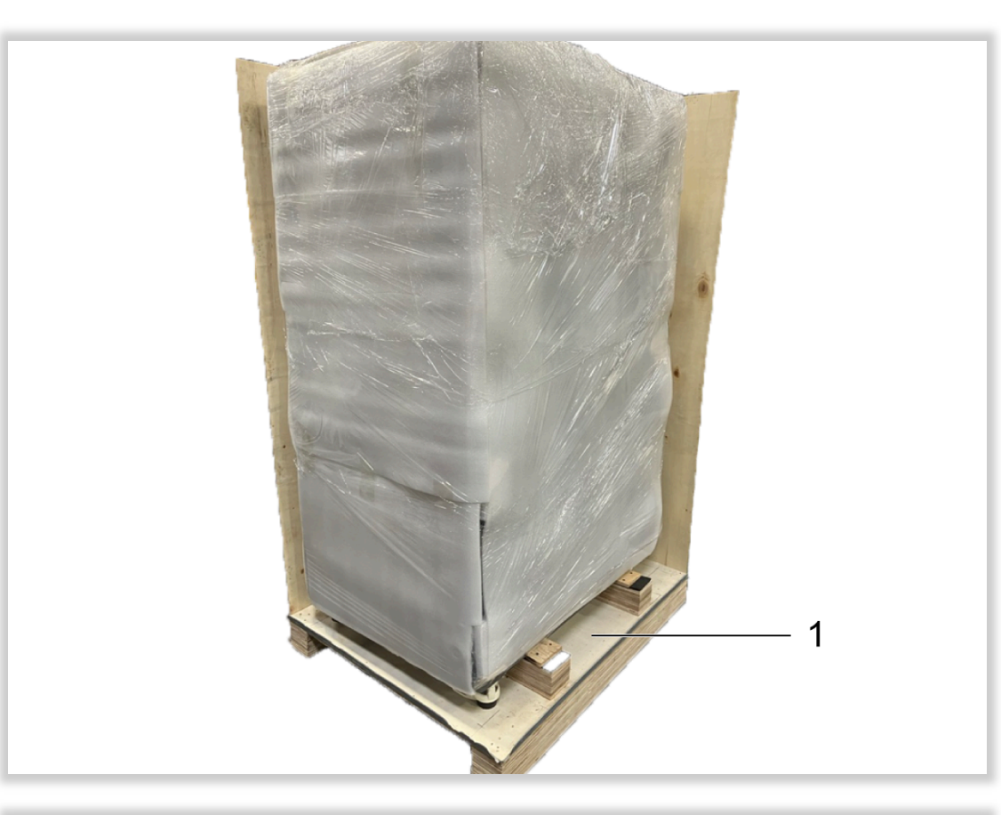
⚠ **IMPORTANT: The Haas laser marker includes a lifetime license to LightBurn Pro Laser Control Software. Upon completion of your purchase, a license key to activate the LightBurn software will be sent to the email associated with your account. If you do not receive an email with your license key, please contact CUSTOMERSUPPORT@HAASCNC.COM. DO NOT activate the trial version of the LightBurn software. LightBurn licenses are perpetual, meaning the software works forever, but the included 365-day update period expires after one year. To continue receiving the newest versions after that year, the customer must purchase a new license.**

Pre-Installation

This system should be installed in an environment that meets the following requirements:

- Smoke and dust-free
- No exposure to acid vapor or other corrosive gases
- Flat foundation

Laser Unpacking

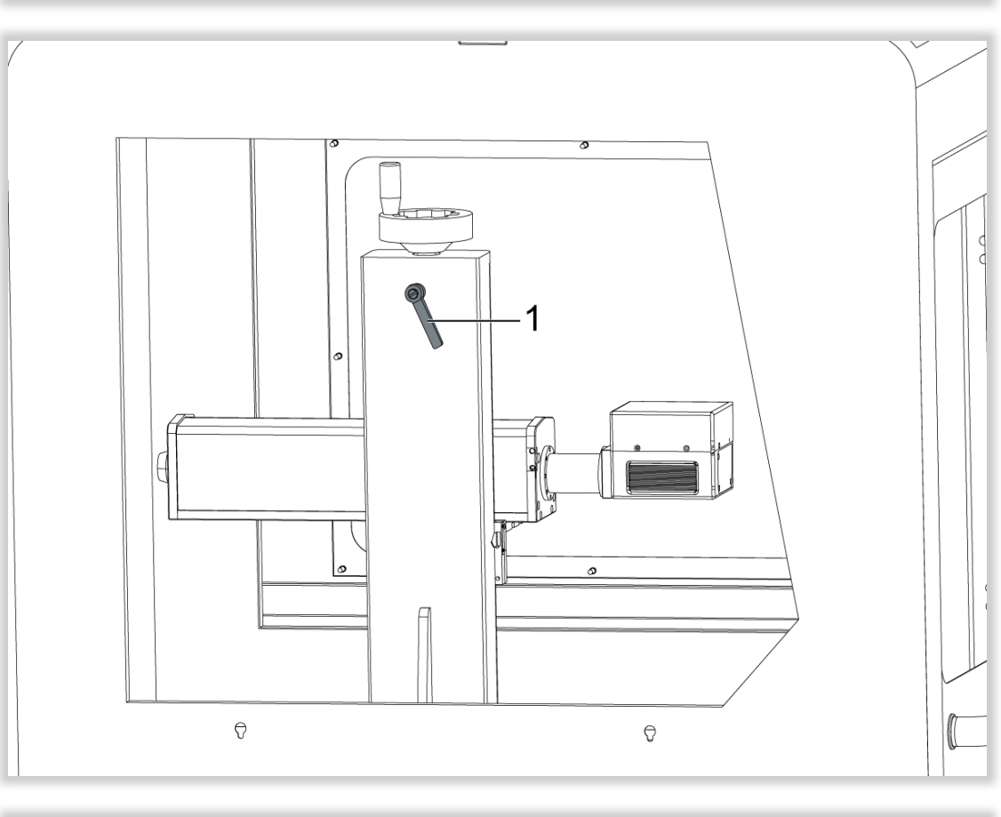


1
Only two side panels and the top panel from the crate need to be removed as shown in the image.

Move the laser to the desired location and remove the wrapping.

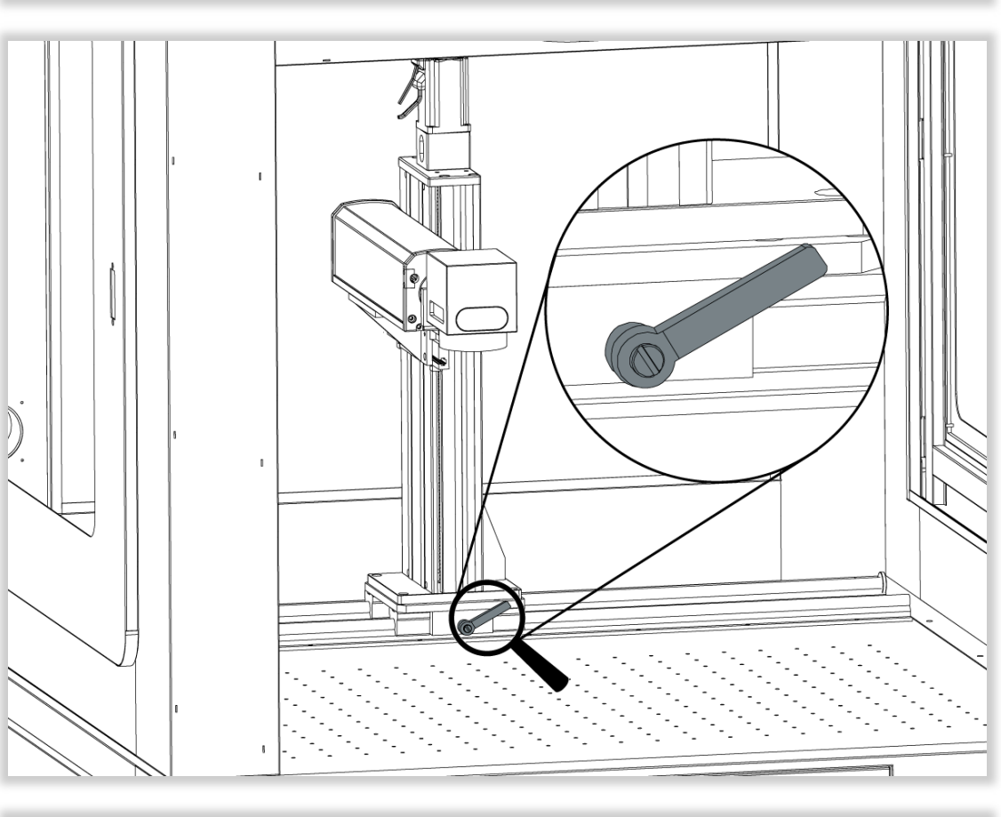
A forklift will be needed to remove the laser from the pallet [1].

The laser machine can be easily moved after removing the brake pads found on each wheel.



2
HL-50E ONLY

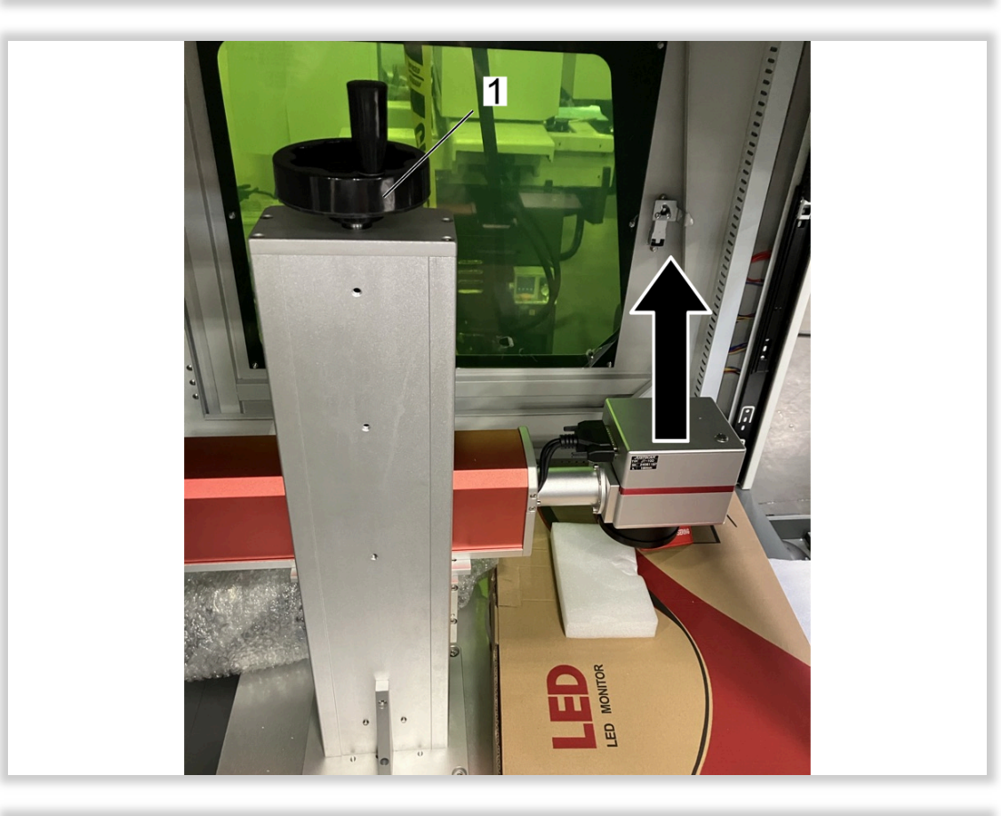
Loosen the Handle Screw to the Z-axis [1].



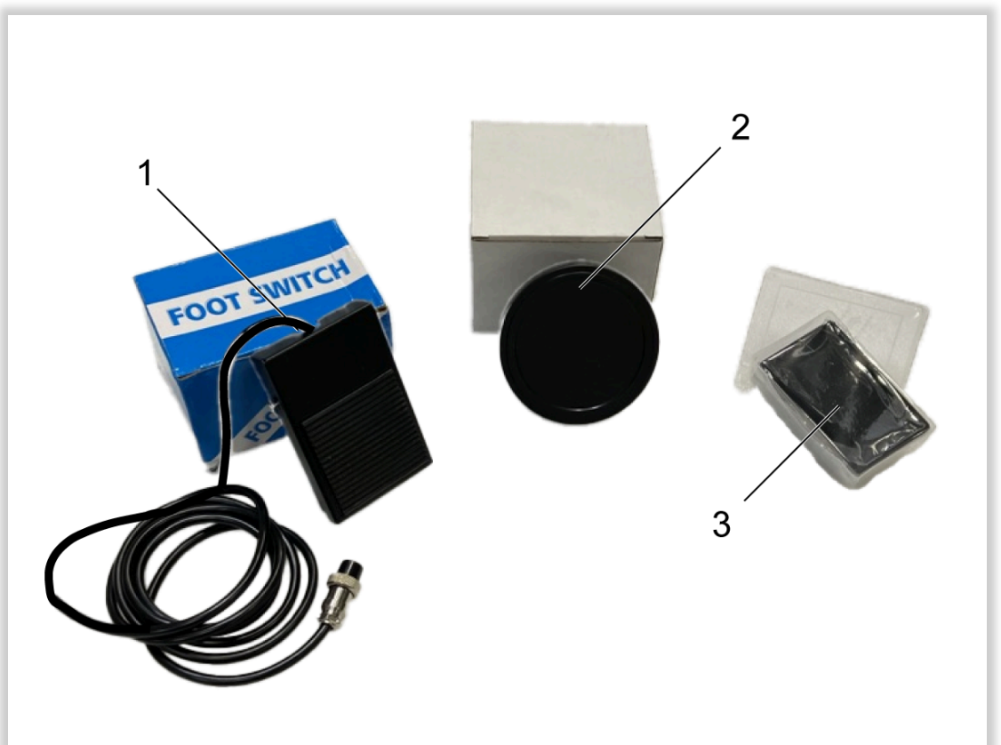
3
HL-50E-XT ONLY

Loosen the X-axis handle screw [1] to allow the laser head to move side to side along the X-axis. Before running a project, tighten the screw as this acts as a stop to keep the laser head from moving during operation.

⚠ **WARNING: Do not actuate or displace the laser head along the X-axis during active laser operation, as this poses significant safety risks and may result in equipment damage or personal injury. Ensure that all side access panels and the main enclosure door remain securely closed while the laser is in use to maintain proper containment and interlock functionality.**



4
With either side window opened, use the laser handle [1] to manually move the laser head upwards for unobstructed access to the monitor box.



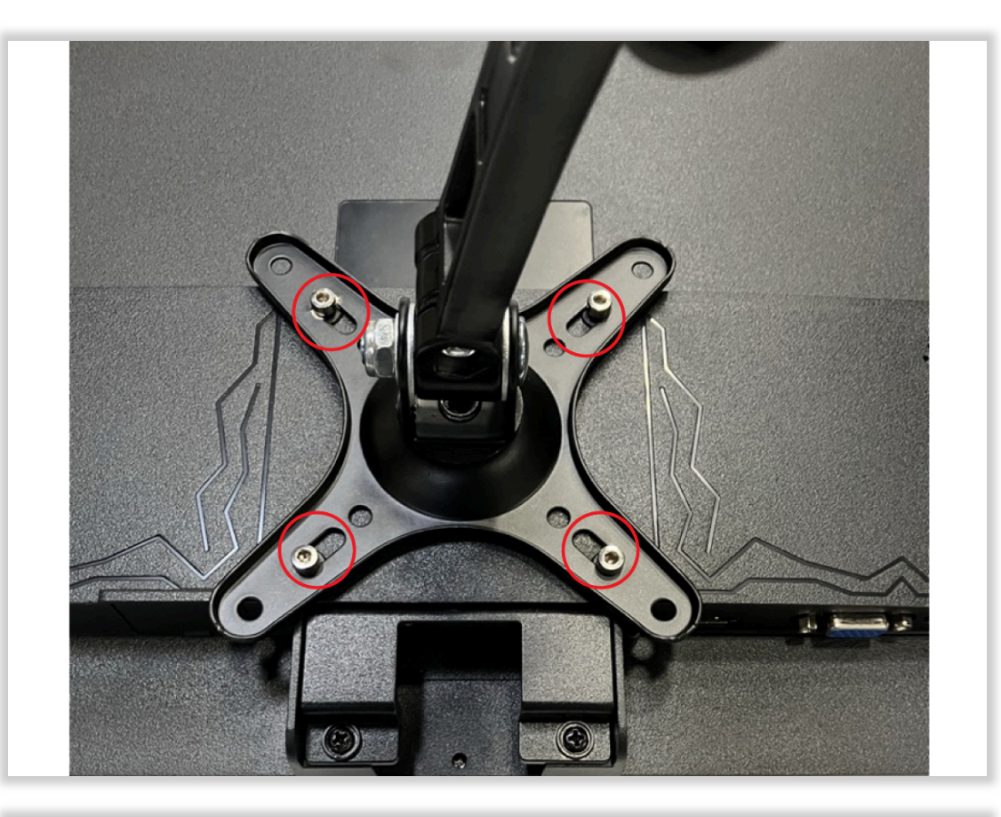
5
Included accessories:

Foot Switch [1]

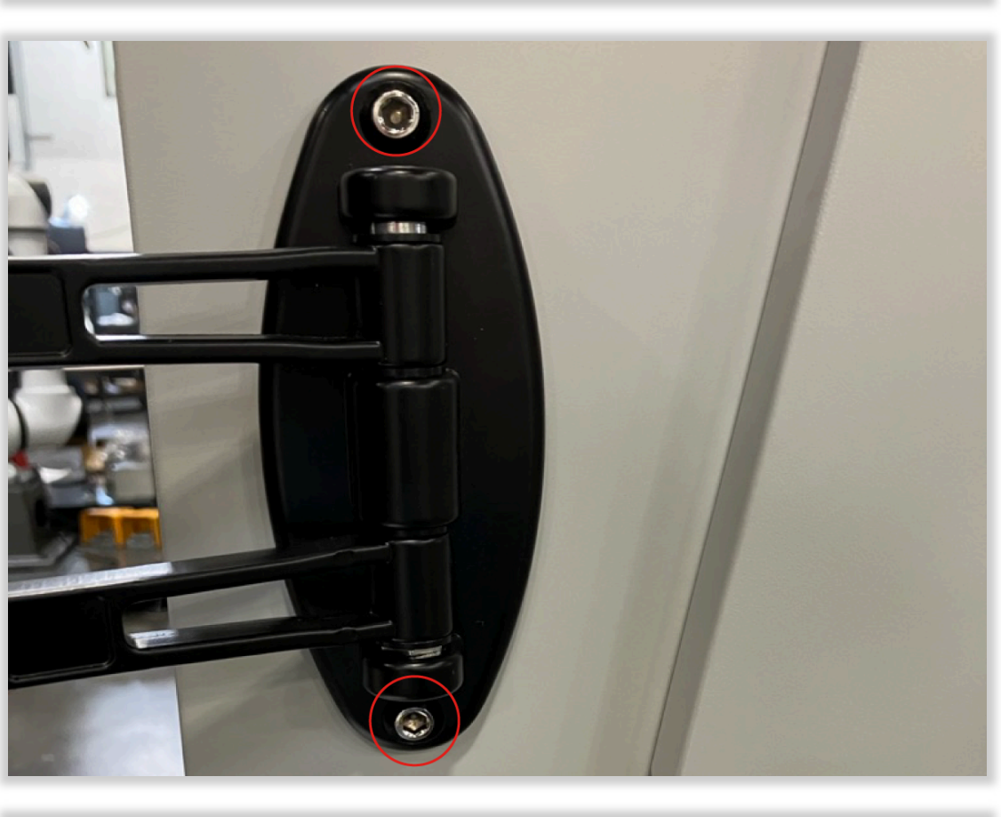
F290 [210x210mm] F-Theta Lens [2]

Anodized Aluminum Blank Cards [3]

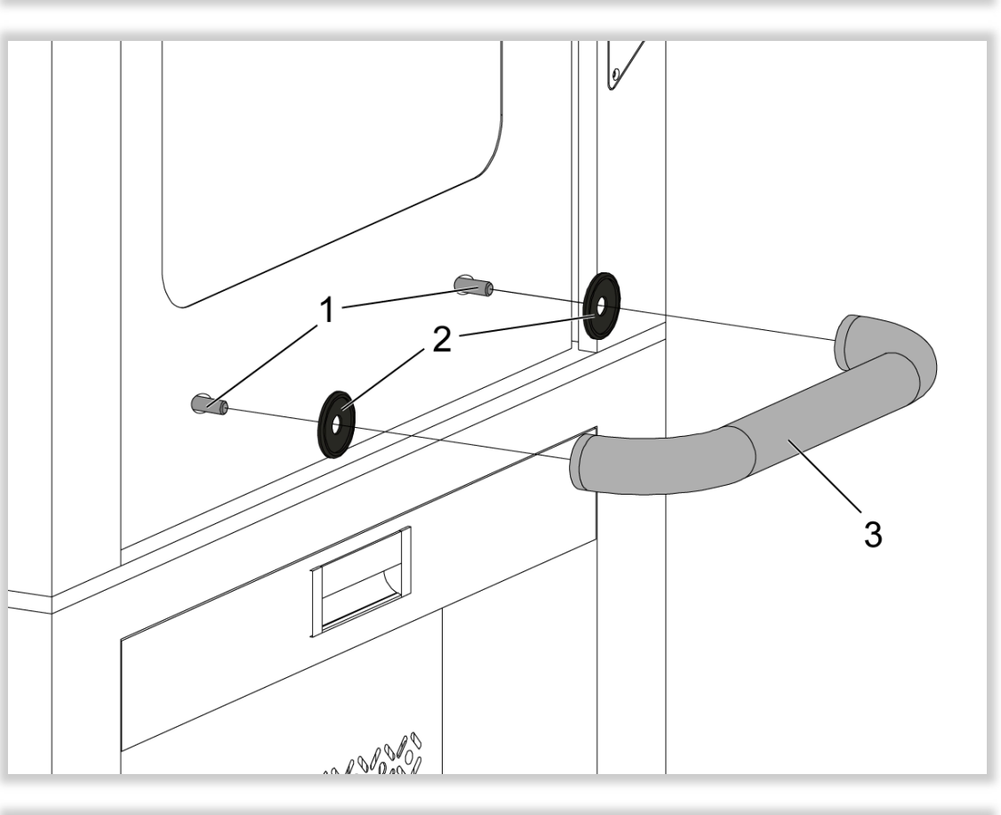
Installation



1
Screw on the mounting arm assembly to the monitor using the (4) bolts using the hexkey set that was provided with the laser marker.



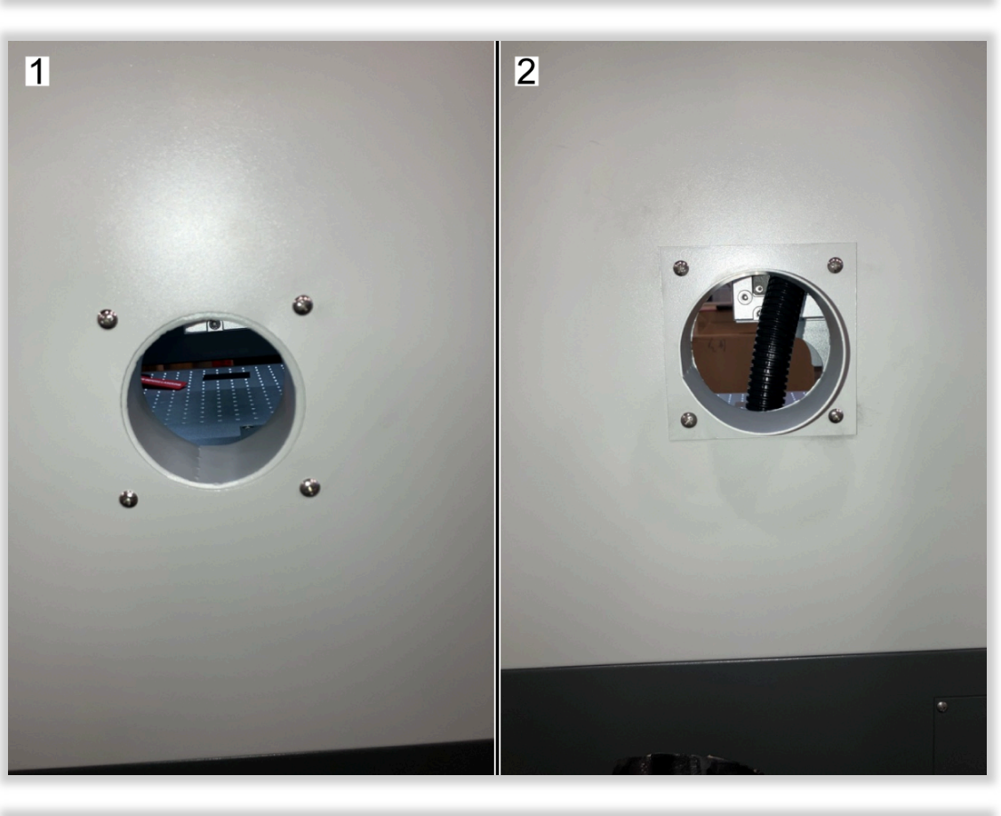
2
Install the monitor mount using the (2) bolts located on right side of the laser machine enclosure.



3
Install the handle by accessing the side window to feed the (2) bolts [1] through the front door as shown in the image.

Add the ferrules [2] before screwing on the handle [3].

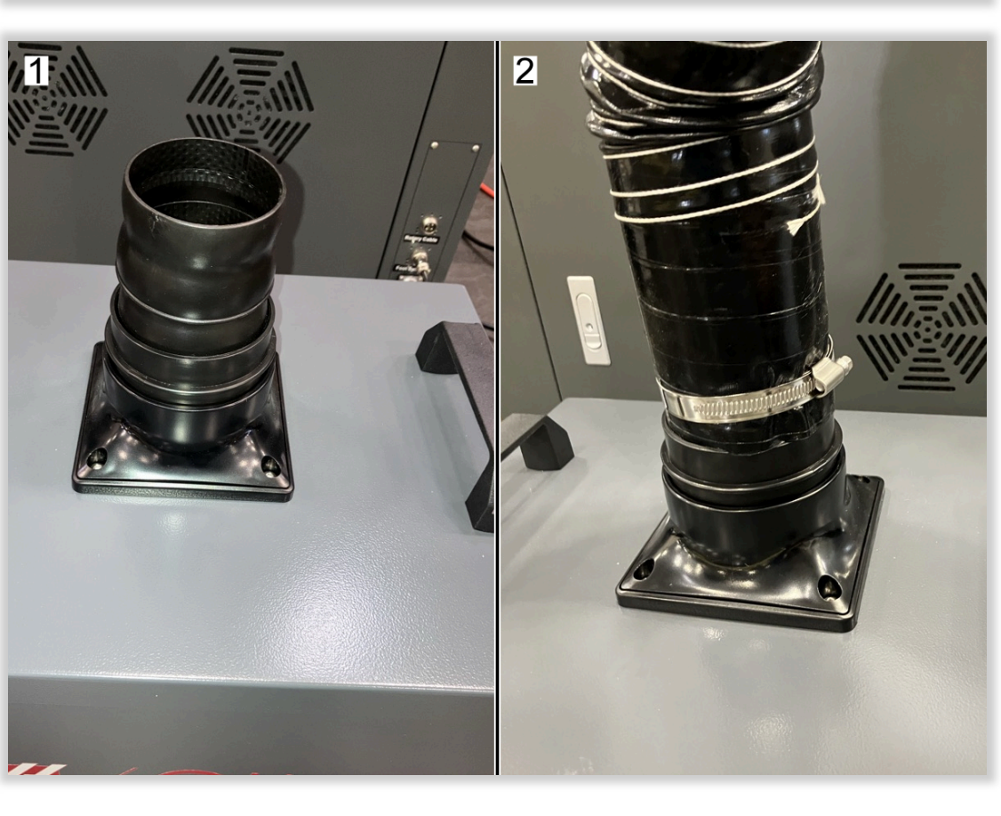
NOTE: Ferrules are installed in a specific orientation. The flat side should be sitting flush against the front door.



4
Invert the air duct bracket [1] so that it is now installed on the outside back panel of the laser machine as shown in image [2].

The cylinder should be facing outward to connect to the fume extractor air duct, using the included air duct clamps.

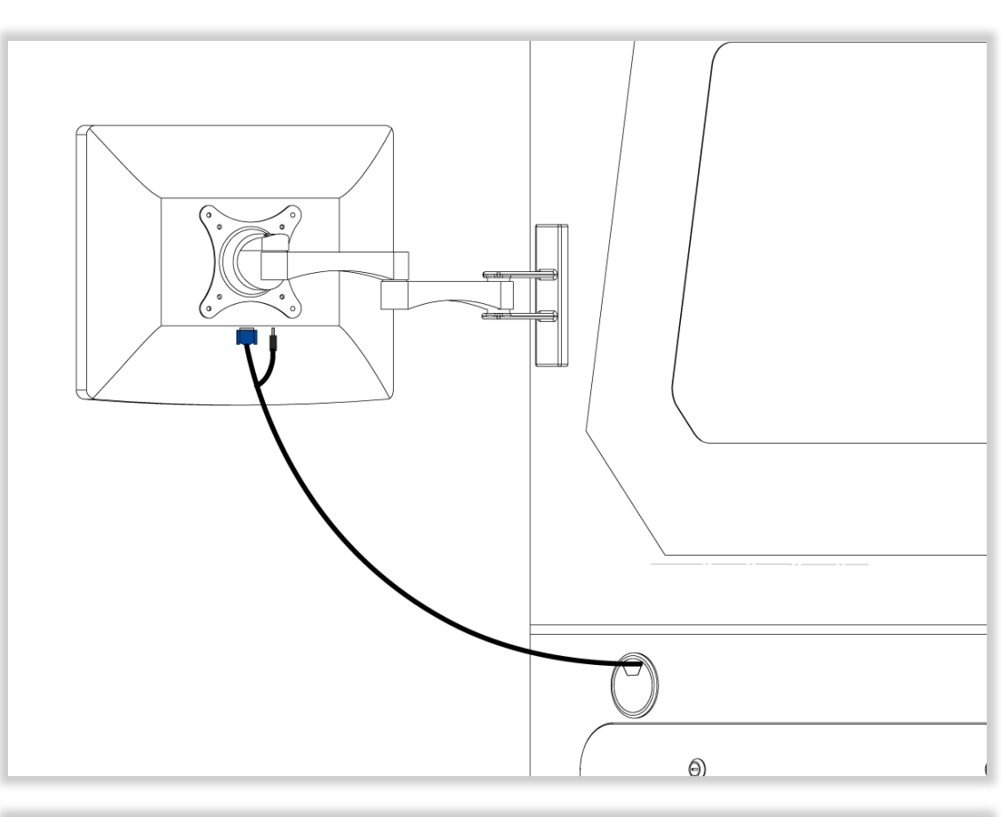
5
Unbox the fume extractor.



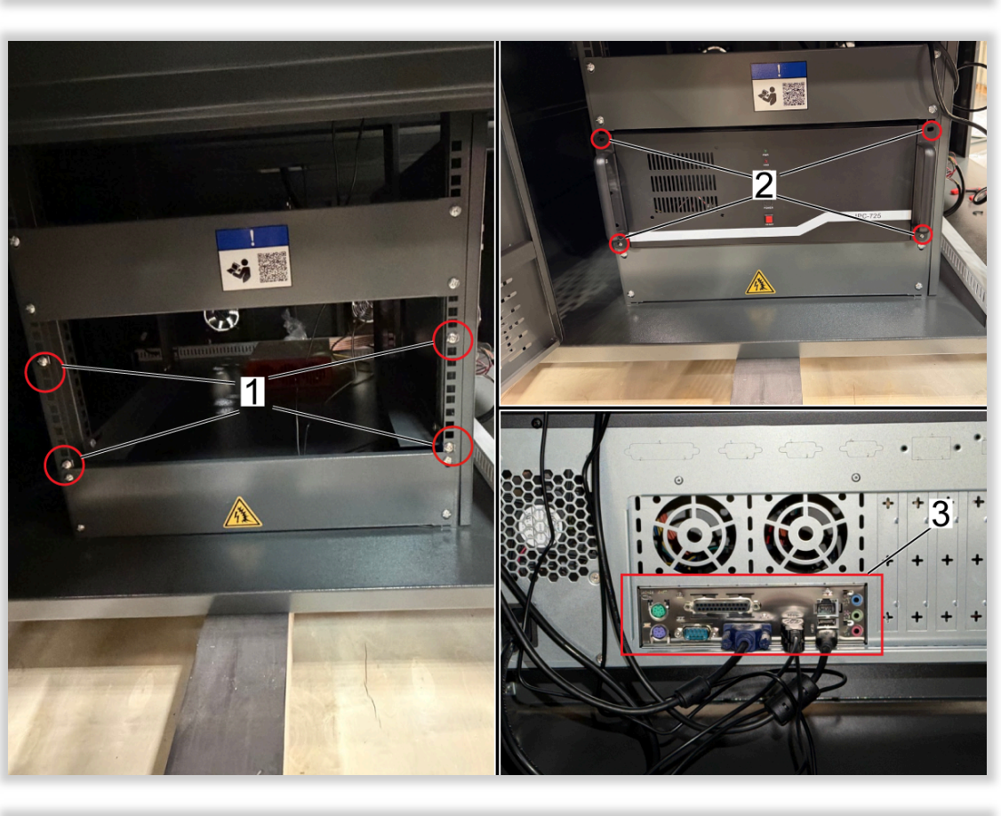
6
Attach the elbow duct fitting [1] to the fume extractor.

Connect the air duct to the elbow duct fitting using the included air duct clamps [2].

Electrical Installation



1
Connect the power/signal cable to the monitor.

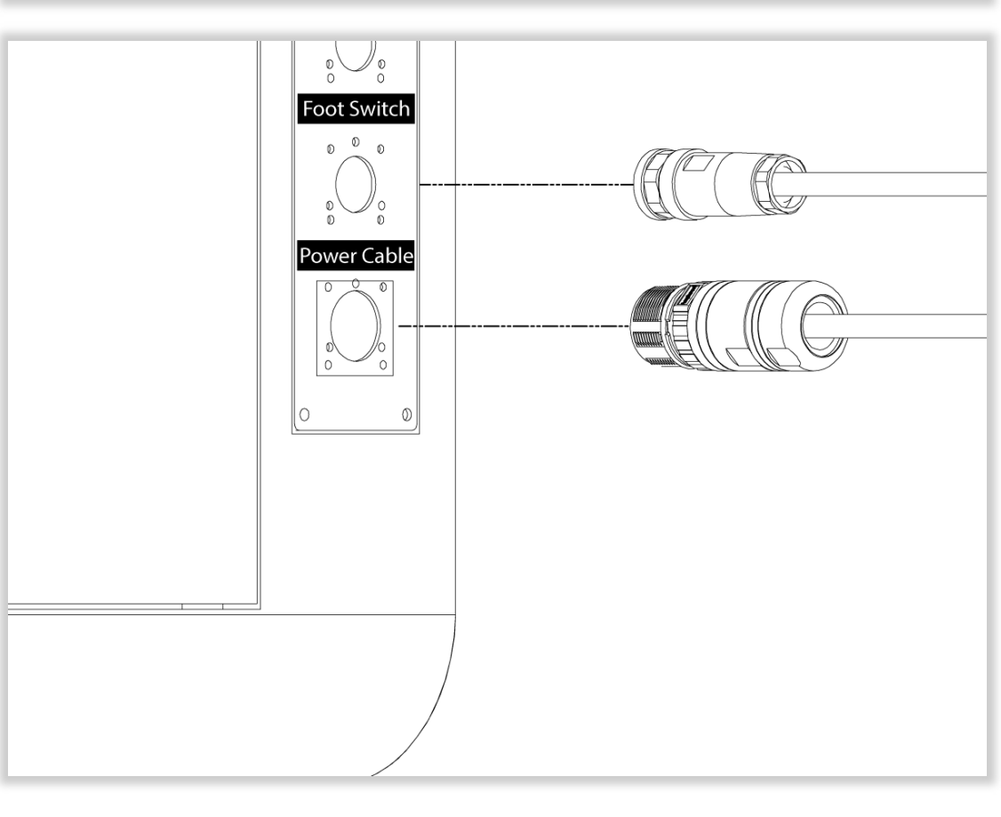


2
Note: This step is for HL-50E-XT or HL-50E machines that shipped after 1/08/2026.

Open the front access door to the PC and remove the (4) screws [1] so they can be used to install the PC.

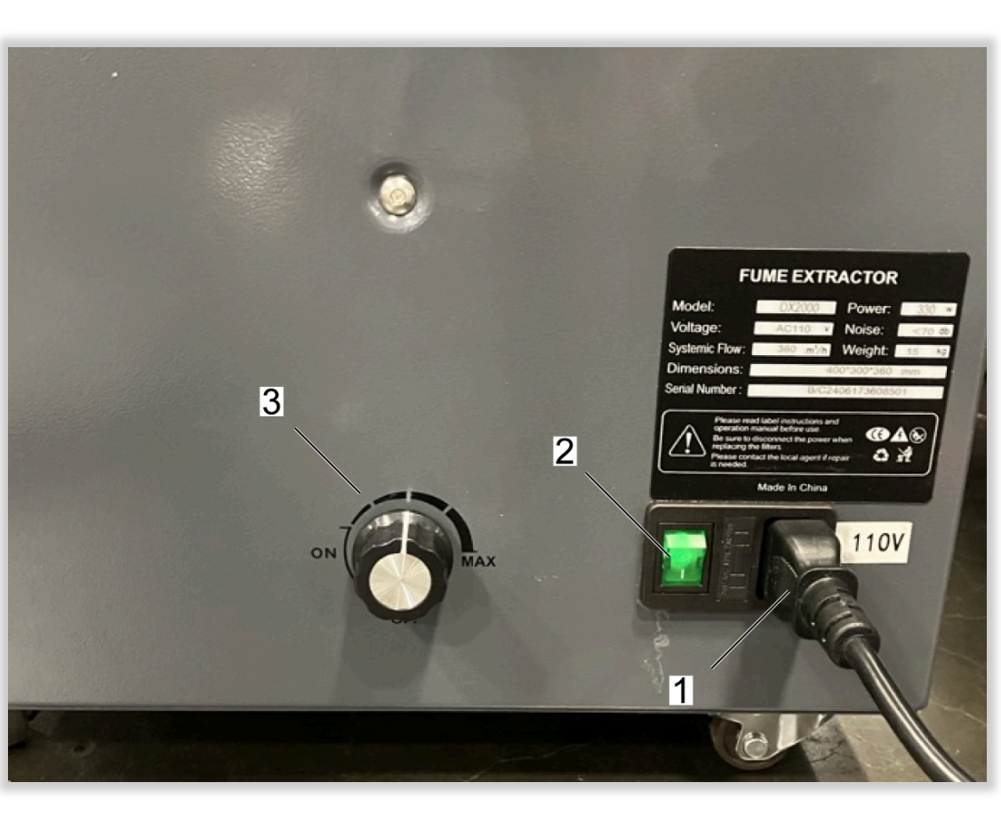
Slide in the PC and use the (4) screws [2] to secure to the PC mount.

Connect the mouse, keyboard, power, laser source signal cable, and monitor signal cable to the PC [3].



3
Connect the foot switch and power cable located in the back of the laser machine.

Plug the power cable to an outlet.



4
Connect the female side of the power cable [1] to the fume extractor and connect the male side to an outlet.

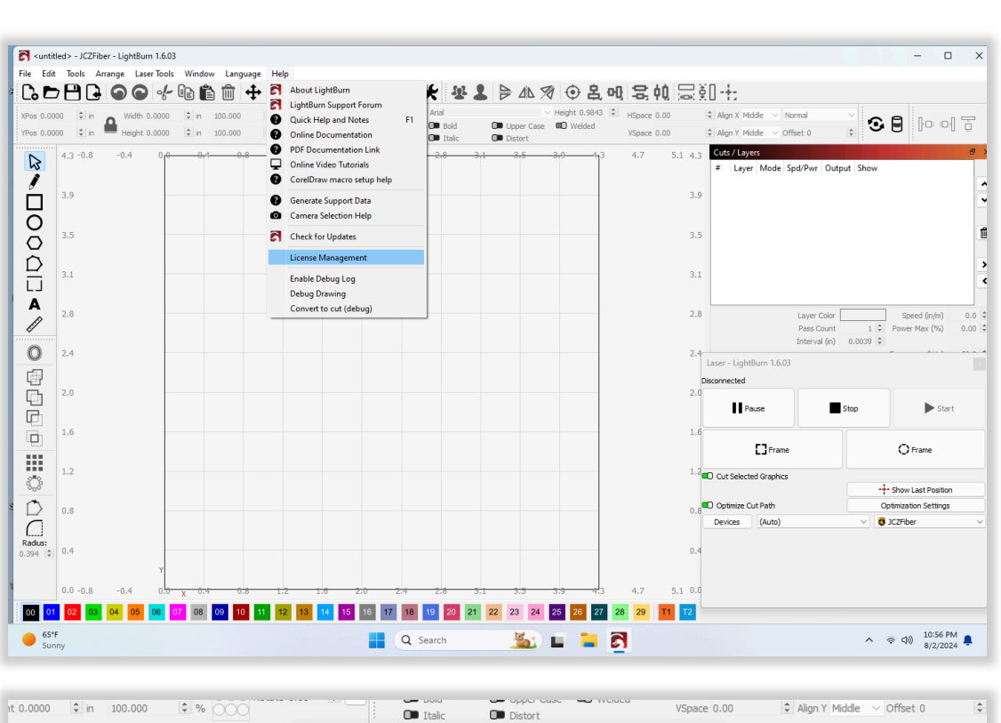
Use switch [2] to turn the fume extractor on or off.

The fume extractor knob [3] can be used to adjust the intensity of the vacuum.

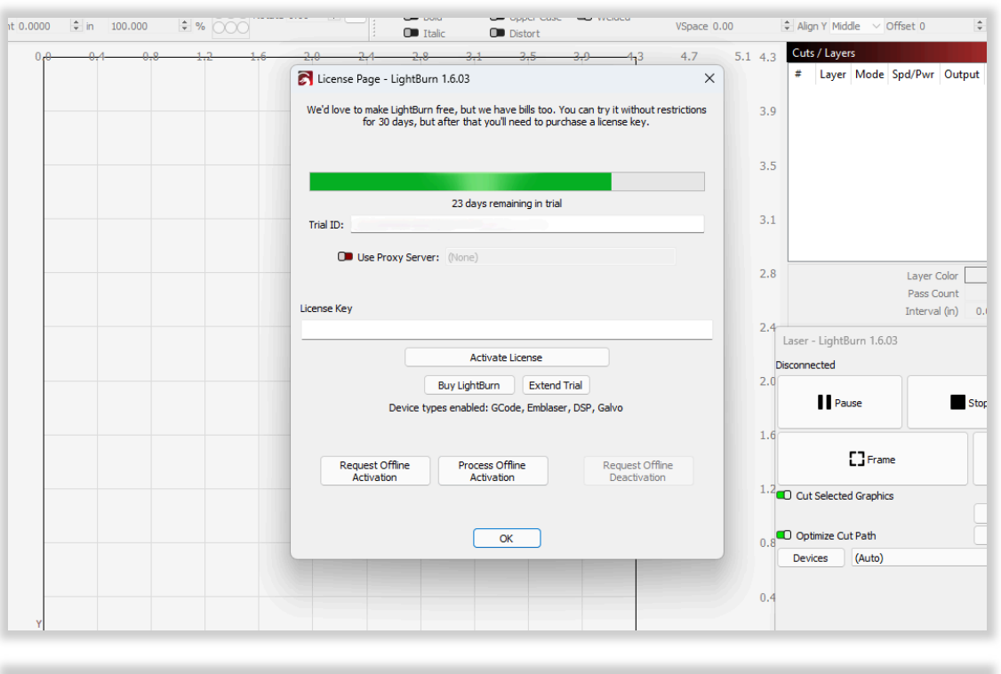
Lightburn License Activation

⚠ **IMPORTANT: Internet will be needed to complete the Lightburn installation.**

⚠ **IMPORTANT: The Haas laser marker includes a lifetime license to LightBurn Pro Laser Control Software. Upon completion of your purchase, a license key to activate the LightBurn software will be sent to the email associated with your account. If you do not receive an email with your license key, please contact CUSTOMERSUPPORT@HAASCNC.COM. DO NOT activate the trial version of the LightBurn software. LightBurn licenses are perpetual, meaning the software works forever, but the included 365-day update period expires after one year. To continue receiving the newest versions after that year, the customer must purchase a new license.**



1
Hover over the **Help** tab and navigate to **License Management** and select it.


License Page - LightBurn 1.6.03
Trial ID:
Use Proxy Server:
License Key
Activate License
Buy LightBurn
Extend Trial
Request Offline Activation
Process Offline Activation
Request Offline Deactivation
OK

We'd love to make LightBurn free, but we have bills too. You can try it without restrictions for 30 days, but after that you'll need to purchase a license key.

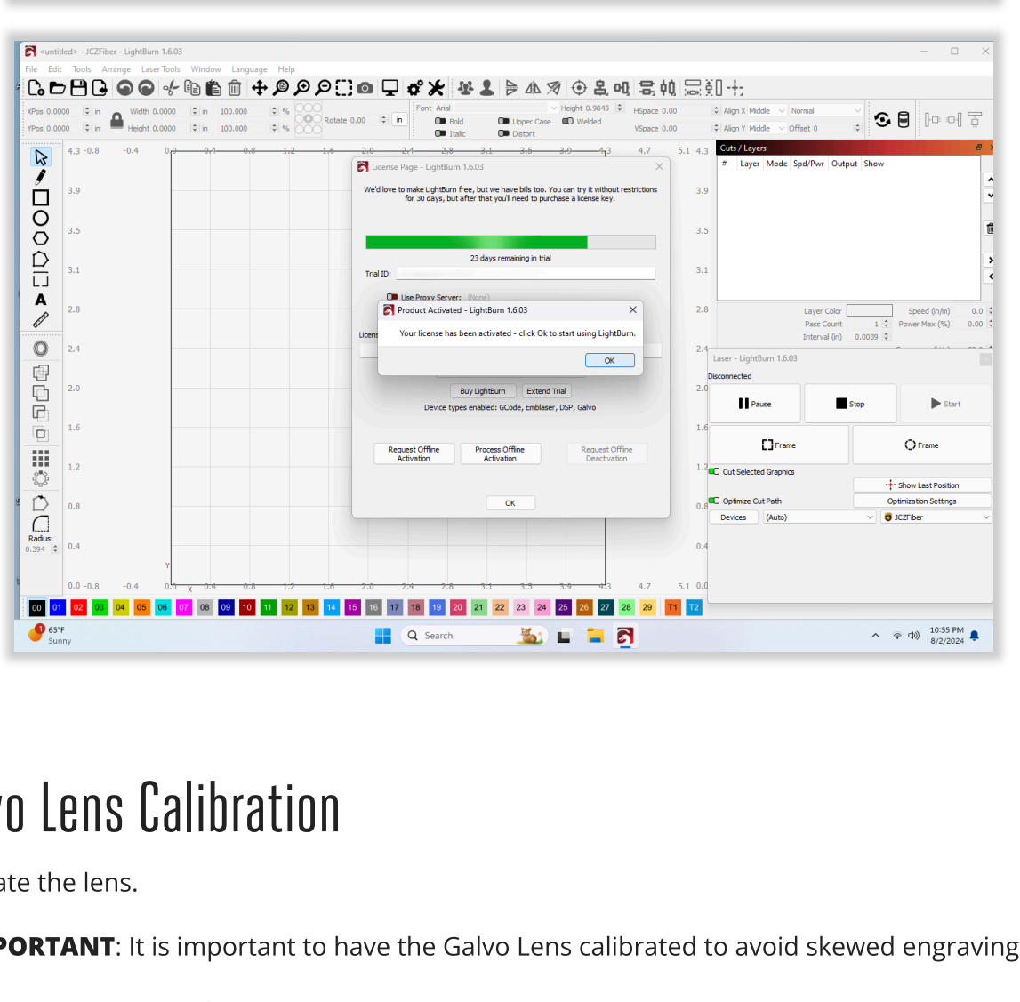
23 days remaining in trial

Device types enabled: GCode, Emblaser, DSP, Galvo

2
Enter the **License Key**.
NOTE: When entering the **License Key**, be sure to include the hyphen(-).

Click **Activate License**.

Click **Ok.**



Your license has been activated - click Ok to start using LightBurn.

3
Click **Ok.**

⚠ **IMPORTANT: If you find that the laser is not connecting to Lighburn, refer to the Laser Drivers section in the HAAS LASER MARKER - TROUBLESHOOTING GUIDE.**

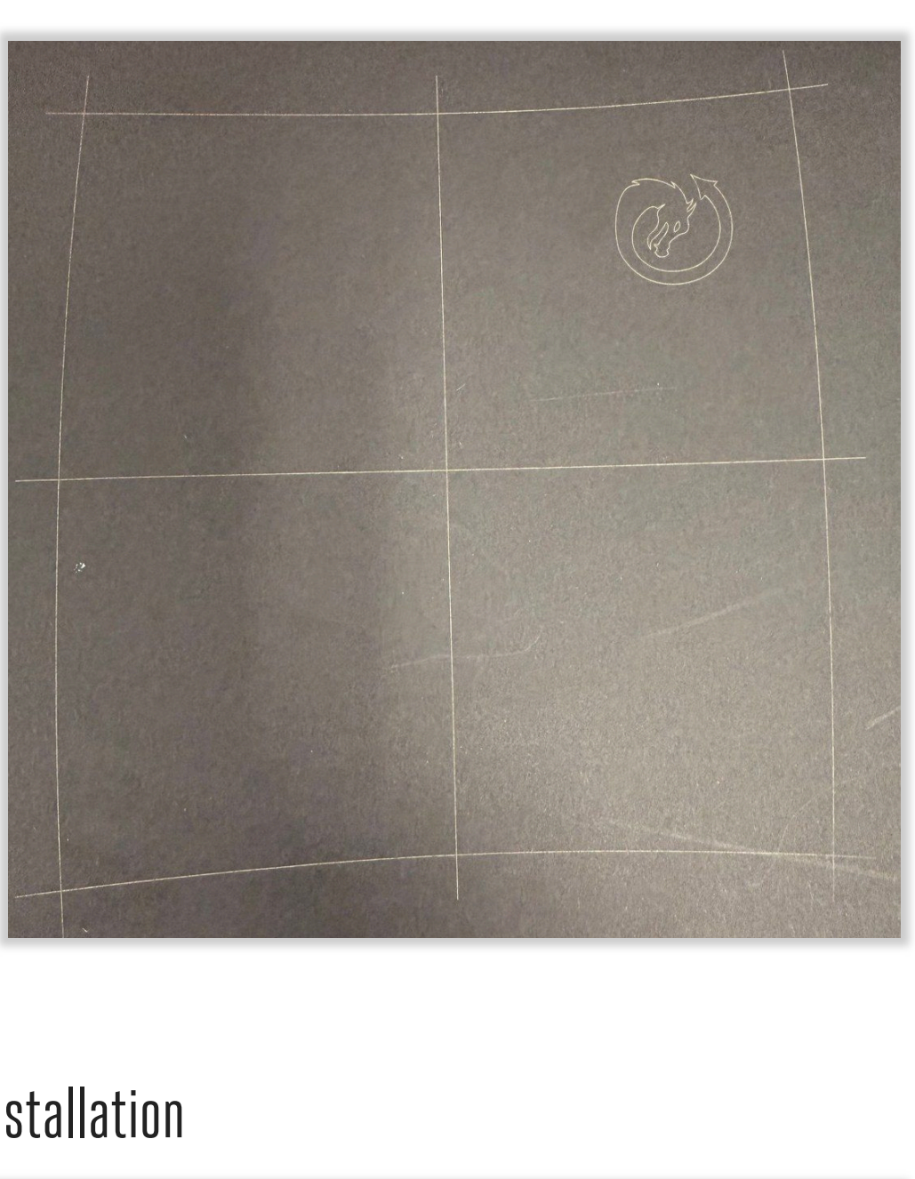
Galvo Lens Calibration

Calibrate the lens.

⚠ **IMPORTANT**: It is important to have the Galvo Lens calibrated to avoid skewed engravings and any bulging on the material. Remove lens caps before operating laser.

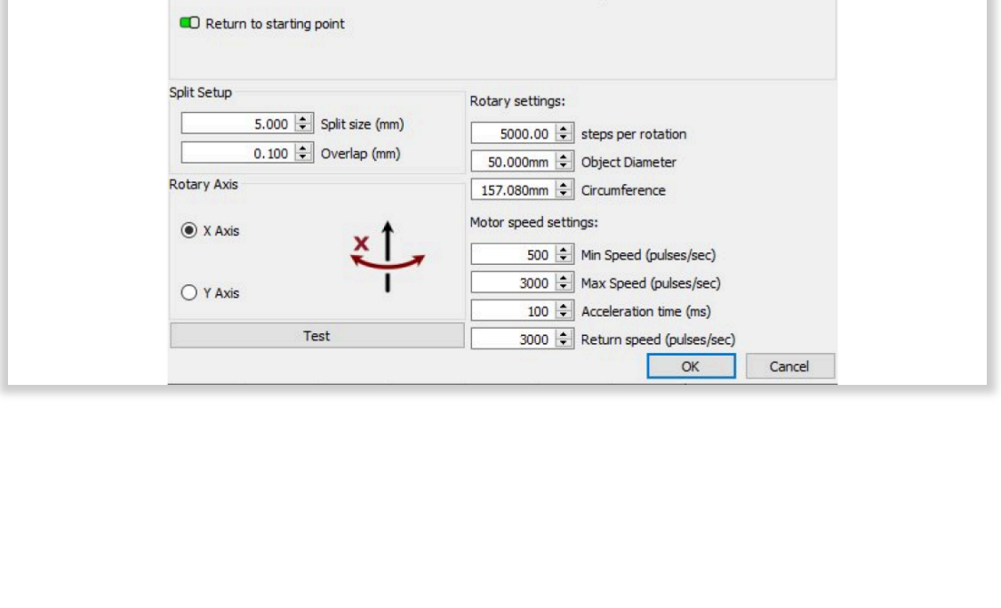
To calibrate the lens, refer to GALVO LENS CALIBRATION.

You can also find relevant information in the Swapping Lenses section in the Operations Chapter found in the manual, see: 4 - HL50E SERIES - OPERATION



This image illustrates the expected reference markings on the black construction paper following execution of the Galvo lens calibration procedure.

Rotary Installation



● Chuck
5.000 Split size (mm)
0.100 Overlap (mm)
5000.00 steps per rotation
50.000mm Object Diameter
157.080mm Circumference
● X Axis
500 Min Speed (pulses/sec)
3000 Max Speed (pulses/sec)
100 Acceleration time (ms)
3000 Return speed (pulses/sec)

1
Rotary Mode Operation

After plugging in the rotary unit to the 4-pin aviation cable located inside near the back of the fixture plate labeled **'Rotary Cable'**, open the **Rotary Setup Window** by clicking **Laser Tools > Rotary Setup** or click on the **'Rotary Icon'** on the top of the screen. Click **'Enable Rotary'**.

Steps Per Rotation:

- Depends on motor and drivers
- Most common value for Haas rotaries is 12,800 steps per rotation

Object Diameter:

- Using calipers, measure the diameter of the part you will be marking
- Input the respective object diameter value of your workpiece
- **'Circumference'** is automatically calculated if **'Object Diameter'** value is inputted

Circumference:

- Input the respective circumference value of your workpiece
- Conversely, **'Object Diameter'** will be automatically calculated if the **'Circumference'** value is inputted

Rotary Axis:

- X-axis if work area is from the top to bottom (or vice versa). Also, if rotary is faced vertically.
- Y-axis if work area is from left to right (or vice versa). Also, if rotary is faced horizontally.

Feedback